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Dr. Leila Yarmohammadi, Ph.D.

Power Electronics Researcher | University Lecturer | Senior Electrical Engineer

Research Focus: Intelligent Power Electronics, Wireless Power Transfer, and AI-Driven
Energy Systems

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Contact

 Email: yarmohammadi.ly@gmail.com

 Tehran, Iran

ORCID: 0000-0002-7284-7054

Google Scholar: <https://scholar.google.com/citations?user=FgX3jBAAAAAJ>

LinkedIn: <https://www.linkedin.com/in/leila-yarmohammadi-phd-707a6553>

PROFESSIONAL PROFILE

Power Electronics researcher, university lecturer, and senior electrical engineer with expertise in wireless power transfer (WPT), resonant power converters, intelligent energy systems, and smart infrastructure. My research combines analytical modeling, optimization, experimental validation, and engineering implementation to develop efficient and reliable power electronic systems. I proposed a novel load-independent output current methodology for WPT systems based on immittance matrix theory, enabling simplified converter design while reducing component stress and improving system performance. In parallel with my academic research, I have contributed to the design, supervision, and commissioning of major governmental infrastructure projects, including hospitals and smart public buildings. My current research interests focus on integrating artificial intelligence with power electronics to develop intelligent wireless charging technologies, resilient energy systems, and autonomous electric transportation. My long-term goal is to develop intelligent power electronic systems by integrating artificial intelligence, advanced converter technologies, and wireless power transfer for future sustainable energy infrastructures.

RESEARCH INTERESTS

Wireless Power Transfer | Power Electronics | Artificial Intelligence | Electric Vehicles | Smart Grid | Energy Management | Optimization | Renewable Energy

RESEARCH HIGHLIGHTS

- ✓ Designed and experimentally implemented a high-order resonant converter for wireless power transfer systems.
- ✓ Developed a novel load-independent output current methodology based on immittance matrix theory.
- ✓ Proposed an optimization framework based on active-to-reactive power ratio for resonant converters.
- ✓ Investigated compensation topologies and transformer misalignment in WPT systems.
- ✓ Published research in international peer-reviewed journals and IEEE conferences.
- ✓ Reviewer for IEEE SPEEDAM 2024.
- ✓ Recipient of multiple Top Researcher Awards from the Ministry of Roads and Urban Development.

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EDUCATION

Ph.D. in Electrical Engineering (Power Electronics)

Islamic Azad University, South Tehran Branch (2017–2023)

Talented Student Program | **GPA: 19.08/20 (4.0/4.0)** | Top Student

Dissertation: Design and Experimental Implementation of a High-Order Resonant Converter with Load-Independent Output Current for Wireless Power Transfer Systems

Supervisor: Prof. Javad Olamaei (Senior Member, IEEE)

M.Sc. in Electrical Engineering (Power)

Islamic Azad University, West Tehran Branch (2011–2013)

GPA:18/20 (3.61/4.0), Top Student.

B.Sc. in Electrical Engineering

Garmsar University

Final Project: FPGA-Based Stepper Motor Driver

TEACHING EXPERIENCE

Year	Course	Lecturer of University
2016– Present	- Power Electronics - Principles of Electricity - Electrical Machines - Electric Circuits - Industrial Electronics - Industrial Automation	South Tehran Branch, Islamic Azad University

PROFESSIONAL EXPERIENCE

Year	Position	Organization	Responsibilities
2004– Present	Head of New Technologies Department	Commissioning Organization for State & Public Building Installations (COBI)- Ministry of Roads and Urban Development	- Supervision of electrical engineering for national governmental projects - Smart building technologies - Building Information Modeling (BIM) - Internet of Things (IoT) infrastructure - Energy auditing based on Iranian National Building Code (Chapter 19) - Renewable energy integration - Technical supervision of hospitals and strategic public buildings
2002–2004	Electrical Engineer	Tehran Electric Co. (Medical Equipment)	R&D Department

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Selected Research Contributions

- Developed a load-independent output current methodology for wireless power transfer systems.
- Designed and experimentally validated a high-order resonant converter.
- Proposed an optimization criterion based on the active-to-reactive power ratio.
- Investigated compensation topologies and transformer misalignment for WPT systems.
- Integrated research outcomes with practical engineering implementation in smart infrastructure projects.

SELECTED PUBLICATIONS

- [1] Yarmohammadi, L. et al. Machine Learning-Driven Energy Management for Critical Load Prioritization During Extreme Events. NAPS 2025.
- [2] Yarmohammadi, L., Siadatan, A., Afjei, E. Analysis of Magnetic Performance of EV Charging Coil Structures. SPEEDAM, Italy, 2024 (Oral Presentation).
- [3] Yarmohammadi, L.; Hosseini, S.M.H.; Olamaei, J.; Mozafari, B. “A Load-Independent Output Current Method for Wireless Power Transfer Systems with Optimal Parameter Tuning”. Sustainability 2022, 14, 9391. [DOI: 10.3390/su14159391](https://doi.org/10.3390/su14159391)
- [4] Yarmohammadi, L.; Hosseini, S.M.H.; Olamaei, J.; Mozafari, B. “Maximizing the efficiency of LCLCL-T Compensated Wireless Power Transfer Systems with independent load Characteristics for charging applications in Smart Home”. Journal of control (English edition), vol 1, issue 2, 2022. [DOI: 10.61186/joc.17.3.27](https://doi.org/10.61186/joc.17.3.27)
- [5] Hamed rashidi aghdam, Leila yarmohammadi, Hosein malakooti, “Studying variety of intelligent control system techniques in hospitals for optimization of energy consumption”, Journal management system, vol 11, issue 21, autumn 2018.
- [6] Leila Yarmohammadi, Mohammad Hassan Hosseini, “Damping of electromagnetic disturbances transient in HTS transmission system”, 2th International Conference on Electrical Engineering, May 2017. (Top Paper).
- [7] L.Yarmohammadi, E.Afjei, A.Siadatan, “Structure novel Boost topology for producing high quality supply for HB-LEDs”, Research journal of recent science, Vol. 4, issue 9, August 2015.
- [8] L.Yarmohammadi, E.Afjei, A.Siadatan, “Design and construction of HB-LED power supply by novel structure for achievement high power factor”, The 22nd Iranian Conference on Electrical Engineering (ICEE 2014).

Google Scholar contains the complete publication list.

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HONORS & AWARDS

- ✓ Top Researcher Award, Ministry of Roads and Urban Development, (2024, 2022, 2014)
- ✓ Reviewer, IEEE SPEEDAM 2024
- ✓ Best Oral Presentation Award, 2nd International Conference on Electrical Engineering (2017)
- ✓ Top national rankings in entrance and employment examinations

PROFESSIONAL ACTIVITIES

- ✓ IEEE SPEEDAM 2024 Reviewer
- ✓ Former IEEE Member

TECHNICAL SKILLS

Programming	Simulation	Optimization
• MATLAB • Python • C++	• Simulink • PSIM • DigSILENT • ANSYS Maxwell • JMAG	• GAMS

RESEARCH METRICS

Google Scholar Citations	18
h-index	3
ORCID	0000-0002-7284-7054

RESEARCH VISION

My long-term research vision is to develop intelligent power electronic systems that combine advanced converter technologies with Artificial Intelligence to improve the efficiency, reliability, and resilience of future energy infrastructures. My goal is to contribute to next-generation wireless power transfer, autonomous electric transportation, and smart energy systems through interdisciplinary research that bridges theoretical innovation with practical engineering implementation.

REFERENCES

References available upon request.